

I) Spring Cloud Bus :

40) You can Refresh all the Instances of all the MicroServices with one Ping as follows

`http://localhost:8100/actuator/bus-refresh`

Working Steps

41) Update BookSearchMS

a) Add below dependency in pom.xml

```
<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-bus-amqp</artifactId>
</dependency>
```

b) Mark the MicroService with @RefreshScope

@RestController

@RefreshScope

```
public class BookSearchController { }
```

c) Update MyBookSearchMS.properties

#management.security.enabled=false

management.endpoints.web.exposure.include=*

42) Update BookStoreWeb

a) Add below dependency in pom.xml

```
<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-bus-amqp</artifactId>
</dependency>
```

b) Mark the MicroService with @RefreshScope.

@RestController

@RefreshScope

```
public class BookStoreController {
}
```

- c) Update MyBookStoreWeb.properties
- ```
#management.security.enabled=false
management.endpoints.web.exposure.include=*
```

**43) Make sure that Rabbit MQ is Started and Running**

<http://localhost:15672/>

**44) Start all the Apps**

Hit This URL : <http://localhost:5000/mybook>

**45) Change the Property in MyBookSearchMS.properties**

```
jlc.search.message = BookSearch - Default Message - Update 3.0
```

**46) Change the Property in MyBookStoreWeb.properties**

```
jlc.store.message = BookSearch - Default Message – Update 2.0
```

**47) Run the following Git Commands**

```
git status
```

```
git add .
```

```
git commit -m "Update #4 by Srinivas Dande"
```

**48) Hit This URL : <http://localhost:5000/mybook>**

You will not see the Updations Reflected to All the Instaces.

**49) Send the Post Request to Any one of the Instances (Postman)**

<http://localhost:8100/actuator/bus-refresh>

**50) Hit This URL : <http://localhost:5000/mybook>**

You will see the Updations Reflected to All the Instaces.



## **J) Creating Profiles:**

### **Consider the following 4 Environments**

|                    |               |
|--------------------|---------------|
| <b>Development</b> | <b>- dev</b>  |
| <b>QA</b>          | <b>- qa</b>   |
| <b>Staging</b>     | <b>-stage</b> |
| <b>Production</b>  | <b>-prod</b>  |

### **Working Steps:**

#### **51) Create property files for 4 Environments for BookSearchMS as follows**

MyBookSearchMS.properties (Default Profile)

MyBookSearchMS-dev.properties (Dev Profile)

MyBookSearchMS-qa.properties ( QA Profile)

MyBookSearchMS-stage.properties (Staging Profile)

MyBookSearchMS-prod.properties (Prod Profile)

#### **52) Create property files for 4 Environments for BookStoreWeb as follows**

MyBookStoreWeb.properties (Default Profile)

MyBookStoreWeb-dev.properties (Dev Profile)

MyBookStoreWeb-qa.properties ( QA Profile)

MyBookStoreWeb-stage.properties (Staging Profile)

MyBookStoreWeb-prod.properties (Prod Profile)

#### **53) Run the following Git Commands.**

git status

git add .

git commit -m "Update #5 by Srinivas Dande"



**54) Start the ConfigServer and Test the following URLs**

<http://localhost:2345/MyBookSearchMS/default/master>  
<http://localhost:2345/MyBookSearchMS/dev/master>  
<http://localhost:2345/MyBookSearchMS/qa/master>  
<http://localhost:2345/MyBookSearchMS/stage/master>  
<http://localhost:2345/MyBookSearchMS/prod/master>

<http://localhost:2345/MyBookStoreWeb/default/master>  
<http://localhost:2345/MyBookStoreWeb/dev/master>  
<http://localhost:2345/MyBookStoreWeb/qa/master>  
<http://localhost:2345/MyBookStoreWeb/stage/master>  
<http://localhost:2345/MyBookStoreWeb/prod/master>

**55) Set the required Active Profile in bootstartp.properties of BookSearchMS**

`spring.profiles.active=dev`

**56) Set the required Active Profile in bootstartp.properties of BookSearchMS**

`spring.profiles.active=prod`

**57) Start RabbitMQ if not started**

**58) Start Eureka Server**

**59) Start BookStoreWeb and BookSearchMS**

**60) Hit This URL : <http://localhost:5000/mybook>**

## **GIT Repository Setup:**

Location : E:/mybookstore/myjlc-repo

### **A) GIT Repository – Files Required**

#### **1) MyBookSearchMS.properties ( Default Profile)**

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
#management.security.enabled=false
management.endpoints.web.exposure.include=*
jlc.search.message = BookSearch - Default Message - Update 1.0
jlc.search.welcome= Welcome to BookSearchMS - Update 1.0
```

#### **2) MyBookSearchMS-dev.properties ( Dev Profile)**

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
#management.security.enabled=false
management.endpoints.web.exposure.include=*

jlc.search.message = BookSearch - Development Env Message - Update 1.0
```

#### **3) MyBookSearchMS-qa.properties( QA Profile)**

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
#management.security.enabled=false
management.endpoints.web.exposure.include=*
jlc.search.message = BookSearch - QA Env Message - Update 1.0
```

#### 4) **MyBookSearchMS-stage.properties (Staging Profile)**

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
#management.security.enabled=false
management.endpoints.web.exposure.include=*
jlc.search.message = BookSearch - Staging Env Message - Update 1.0
```

#### 5) **MyBookSearchMS-prod.properties (Prod Profile)**

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
#management.security.enabled=false
management.endpoints.web.exposure.include=*
jlc.search.message = BookSearch - Production Env Message - Update 1.0
```

#### 6) **MyBookStoreWeb.properties ( Default Profile)**

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
#management.security.enabled=false
management.endpoints.web.exposure.include=*
jlc.store.message = BookStore - Default Message -First Update
jlc.store.welcome= Welcome to BookStoreWeb -First Update
```

### 7) MyBookStoreWeb-dev.properties ( Dev Profile)

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
#management.security.enabled=false
management.endpoints.web.exposure.include=*
jlc.store.message = BookStore - Development Env Message -First Update
```

### 8) MyBookStoreWeb-qa.properties( QA Profile)

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
#management.security.enabled=false
management.endpoints.web.exposure.include=*
jlc.store.message = BookStore - QA Env Message -First Update
```

### 9) MyBookStoreWeb-stage.properties (Staging Profile)

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
#management.security.enabled=false
management.endpoints.web.exposure.include=*
jlc.store.message = BookStore - Staging Env Message -First Update
```

### 10) MyBookStoreWeb-prod.properties (Prod Profile)

```
eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n
#management.security.enabled=false
management.endpoints.web.exposure.include=*
jlc.store.message = BookStore - Production Env Message -First Update
```



## B) MyConfigServer – Files Required

|                                   |                         |
|-----------------------------------|-------------------------|
| 1. MyConfigServerApplication.java | 2. bootstrap.properties |
| 3. pom.xml                        |                         |

### 1. MyConfigServerApplication.java

```
package com.jlcindia.myconfigserver;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.cloud.config.server.EnableConfigServer;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@SpringBootApplication
@EnableConfigServer
public class MyConfigServerApplication {
 public static void main(String[] args) {
 SpringApplication.run(MyConfigServerApplication.class, args);
 }
}
```

### 2. bootstrap.properties

```
spring.application.name=MyConfigServer
server.port=2345
spring.cloud.config.server.git.uri = E:/mybookstore/myjlc-repo
```

### 3. pom.xml

```
<project ...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.myconfigserver</groupId>
<artifactId>MyConfigServer</artifactId>
<version>0.0.1-SNAPSHOT</version>
<name>MyConfigServer</name>
<description>ConfigServer - By Srinivas Dande</description>

<properties>
<java.version>1.8</java.version>
<spring-cloud.version>Hoxton.SR8</spring-cloud.version>
</properties>
```



```
<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.3.3.RELEASE</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-actuator</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-config-server</artifactId>
</dependency>

</dependencies>

<dependencyManagement>
<dependencies>
<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-dependencies</artifactId>
<version>${spring-cloud.version}</version>
<type>pom</type>
<scope>import</scope>
</dependency>
</dependencies>
</dependencyManagement>

<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```



### C) MyEurekaServer – Files Required

4. MyEurekaServerApplication.java	5. application.properties
6. pom.xml	

#### 1. MyEurekaServerApplication.java

```
package com.jlcindia.myeurekaserver;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.cloud.netflix.eureka.server.EnableEurekaServer;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@SpringBootApplication
@EnableEurekaServer
public class MyEurekaServerApplication {
 public static void main(String[] args) {
 SpringApplication.run(MyEurekaServerApplication.class, args);
 }
}
```

#### 2. application.properties

```
spring.application.name=MyEurekaServer
server.port=8761

eureka.client.register-with-eureka=false
eureka.client.fetchRegistry= false
```

#### 3. pom.xml

```
<project ...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.myeurekaserver</groupId>
<artifactId>MyEurekaServer</artifactId>
<version>0.0.1-SNAPSHOT</version>
<name>MyEurekaServer</name>
<description>Eureka - Service Registry and Discovery Server</description>

<properties>
<java.version>1.8</java.version>
<spring-cloud.version>Hoxton.SR8</spring-cloud.version>
</properties>
```



```
<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.3.3.RELEASE</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-actuator</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-netflix-eureka-server</artifactId>
</dependency>

</dependencies>

<dependencyManagement>
<dependencies>
<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-dependencies</artifactId>
<version>${spring-cloud.version}</version>
<type>pom</type>
<scope>import</scope>
</dependency>
</dependencies>
</dependencyManagement>

<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```



## D) BookSearchMS – Files Required

1. BookSearchApplication.java	2. BookSearchController.java
3. BookInfo.java	4. bootstrap.properties
5. pom.xml	

### 1. BookSearchApplication.java

```
package com.jlcindia.booksearch;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.cloud.netflix.eureka.EnableEurekaClient;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@SpringBootApplication
@EnableEurekaClient
public class BookSearchApplication {
 public static void main(String args[]) {
 SpringApplication.run(BookSearchApplication.class, args);
 }
}
```

### 2. BookSearchController.java

```
package com.jlcindia.booksearch;

import org.springframework.beans.factory.annotation.Value;
import org.springframework.cloud.context.config.annotation.RefreshScope;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@RestController
@RefreshScope
public class BookSearchController {

 @Value("${server.port}")
 String searchPort;
```



```
@Value("${jlc.search.message}")
```

```
String searchMessage;
```

```
@GetMapping("/jlcbook")
```

```
public BookInfo getBook() {
```

```
 System.out.println("---BookSearchController---getBook()---");
```

```
 BookInfo bookInfo=new BookInfo();
```

```
 bookInfo.setBookId(101);
```

```
 bookInfo.setBookName("Master Spring MicroServices");
```

```
 bookInfo.setAuthor("Srinivas Dande");
```

```
 bookInfo.setSearchPort(searchPort);
```

```
 bookInfo.setSearchMessage(searchMessage);
```

```
 return bookInfo;
```

```
}
```

```
}
```

### 3. BookInfo.java

```
package com.jlcindia.booksearch;
```

```
/*
```

```
* @Author : Srinivas Dande
```

```
* @company : Java Learning Center
```

```
**/
```

```
public class BookInfo {
```

```
 private Integer bookId;
```

```
 private String bookName;
```

```
 private String author;
```

```
 private String searchPort;
```

```
 private String searchMessage;
```

```
 //Constructors
```

```
 //Setters and Getters
```

```
}
```

### 4. bootstrap.properties

```
spring.application.name=MyBookSearchMS
```

```
server.port=8000
```

```
spring.cloud.config.uri=http://localhost:2345
```

```
spring.profiles.active=prod
```



## 5. pom.xml

```
<project ...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.booksearch</groupId>
<artifactId>BookSearchMS</artifactId>
<version>1.0.SNAPSHOT</version>
<packaging>jar</packaging>

<name>MyBookSearchMS</name>
<url>https://www.jlcindia.com</url>
<description>This is BookSeach MS developed by Srinivas Dande</description>

<properties>
<java.version>1.8</java.version>
<spring-cloud.version>Hoxton.SR8</spring-cloud.version>
</properties>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.3.3.RELEASE</version>
</parent>

<dependencies>
<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-bus-amqp</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-config</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-actuator</artifactId>
</dependency>
```



```
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

</dependencies>

<dependencyManagement>
<dependencies>
<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-dependencies</artifactId>
<version>${spring-cloud.version}</version>
<type>pom</type>
<scope>import</scope>
</dependency>
</dependencies>
</dependencyManagement>

<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>

</project>
```



## E) BookStoreWeb – Files Required

1. BookStoreApplication.java	2. BookSearchProxy.java
3. BookStoreController.java	4. BookInfo.java
5. bootstrap.properties	6. pom.xml

### 1. BookStoreApplication.java

```
package com.jlcindia.bookstoreweb;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.cloud.netflix.eureka.EnableEurekaClient;
import org.springframework.cloud.openfeign.EnableFeignClients;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@SpringBootApplication
@EnableFeignClients
@EnableEurekaClient
public class BookStoreApplication {
 public static void main(String args[]) {
 SpringApplication.run(BookStoreApplication.class, args);
 }
}
```

### 2. BookSearchProxy.java

```
package com.jlcindia.bookstoreweb;

import org.springframework.cloud.openfeign.FeignClient;
import org.springframework.web.bind.annotation.GetMapping;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@FeignClient("MyBookSearchMS")
public interface BookSearchProxy {

 @GetMapping("/jlcbook")
 public BookInfo getBook();
}
```



### 3. BookStoreController.java

```
package com.jlcindia.bookstoreweb;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.beans.factory.annotation.Value;
import org.springframework.cloud.context.config.annotation.RefreshScope;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

@RestController
@RefreshScope
public class BookStoreController {

 @Autowired
 BookSearchProxy bookSearchProxy;

 @Value("${server.port}")
 String storePort;

 @Value("${jlc.store.message}")
 String storeMessage;

 @GetMapping("/mybook")
 public BookInfo getBook() {
 System.out.println("---BookStoreController---getBook()---");

 BookInfo bookInfo = bookSearchProxy.getBook();

 bookInfo.setStorePort(storePort);
 bookInfo.setStoreMessage(storeMessage);

 return bookInfo;
 }
}
```



#### 4. BookInfo.java

```
package com.jlcindia.bookstoreweb;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
public class BookInfo {
 private Integer bookId;
 private String bookName;
 private String author;
 private String searchPort;
 private String searchMessage;
 private String storePort;
 private String storeMessage;

 //Constructors
 //Setters and Getters
}
```

#### 5. application.properties

```
spring.application.name=MyBookStoreWeb
server.port=5000
spring.cloud.config.uri=http://localhost:2345
spring.profiles.active=dev
```

#### 6. pom.xml

```
<project ...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.bookstoreweb</groupId>
<artifactId>BookStoreWeb</artifactId>
<version>1.3</version>
<packaging>jar</packaging>

<name>MyBookStoreWeb</name>
<url>https://www.jlcindia.com</url>
<description>This is BookStoreWeb developed by Srinivas Dande</description>

<properties>
<java.version>1.8</java.version>
<spring-cloud.version>Hoxton.SR8</spring-cloud.version>
</properties>
```



```
<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.3.3.RELEASE</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-bus-amqp</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-config</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-netflix-eureka-client</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-starter-openfeign</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-actuator</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

</dependencies>
```



```
<dependencyManagement>
<dependencies>
<dependency>
<groupId>org.springframework.cloud</groupId>
<artifactId>spring-cloud-dependencies</artifactId>
<version>${spring-cloud.version}</version>
<type>pom</type>
<scope>import</scope>
</dependency>
</dependencies>
</dependencyManagement>

<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>

</project>
```